

► HUNTING UNDER THE ICE ►

Many seals hunt fish and squid under the ice, surfacing to breathe at the ice edge. But as the Antarctic pack ice expands north in winter, weddell seals stay behind, feeding beneath the pack ice. They breathe through holes that they make in the ice by using their teeth to chip away the ice around existing cracks. Most weddell seals have badly worn teeth as a result, and some die from serious dental infections.



◄ ICY WATERS

Salt seawater freezes at 28.8°F (-1.8°C), so it can be colder than the freezing point of fresh water. Animals that live in the water below the pack ice must cope with temperatures that might freeze their bodies. Many polar fish and shellfish survive because they have natural antifreezes in their body fluids. Since food is plentiful on the seabed, these animals can live beneath the ice in large numbers.



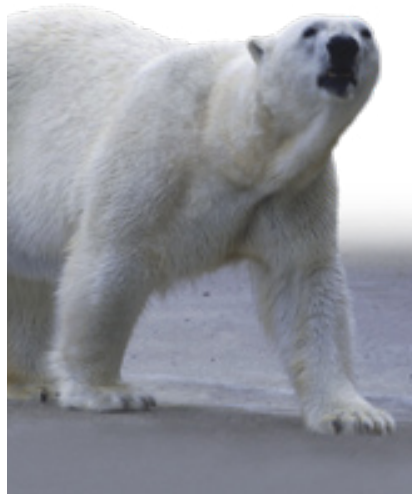
► SEAL COLONIES

In the Arctic, female seals climb out onto the ice to give birth to their pups. Harp seals form large colonies on unstable pack ice, which cannot support the prowling polar bears that are their main enemies. Harp seals feed their white-coated pups on rich milk for just 12 days before returning to the water.



◄ BREEDING PENGUINS

Most Antarctic penguins breed on rocky shores in summer, choosing places where there is no ice to freeze their eggs. Many species like this king penguin breed on the islands fringing the Southern Ocean, which thaw out for longer than the Antarctic continent. It supports its egg on its large webbed feet, covering it with a flap of skin to keep it warm. Uniquely, the closely related emperor penguin uses the same method to breed on sea ice close to Antarctica, incubating its egg throughout the harsh polar winter.



◄ SEA BEAR

The polar bear is a specialized marine predator that lives on the Arctic sea ice. It preys mainly on seals—especially ringed seal pups in spring—and follows the edge of the stable pack ice as it advances and recedes with the seasons. When the pack ice breaks up in summer it is often forced to swim to the land. But here it has trouble finding food, and many polar bears die of starvation. Its future is seriously threatened by the dwindling area of summer sea ice in the Arctic caused by global warming.